

WEEK 7

PHYSICS

Weekly question grade XI

1. Differentiate between dot product (scalar product) and cross product (vector product).
2. Explain the geometrical meaning of the dot product and cross product of two vectors.
3. Find the scalar and vector components of $(\vec{a} + \vec{b})$ in the direction of $(\vec{a} - \vec{b})$.
4. Explain the right-hand thumb rule for the cross product of two vectors.
5. What is the angle between $(\vec{A} \times \vec{B})$ and $(\vec{A} + \vec{B})$? Justify analytically and conceptually.
6. Describe Dulong and Petit's experiment for the determination of the real expansivity of a liquid.
6. Describe Dulong and Petit's experiment for the determination of the real expansivity of a liquid.
7. Define real expansivity and apparent expansivity of a liquid, and find the relation between them.
8. A glass vessel is completely filled with 340 g of mercury at 0°C . What weight of mercury will overflow when the vessel is heated to 100°C ?
(Given: $\gamma_{\text{hg}} = 18 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and α_{glass})
9. State and prove Gauss's theorem of electrostatics.
10. Use Gauss's theorem to find the electric field intensity due to a charged conducting sphere.